



RECOVERY PROCEDURE

VIRTUAL CONTROL
VMWARE CLOUD FOUNDATION SOLUTIONS

vmware-certificatemanagement Service Recovery

When NSX CM register fails with error 90348 "Failed to import the trusted root certificate" because vCenter's cert-mgmt service is stopped. Verified fix from a 2026-05-06 incident.

VCENTER

NSX 9

COMPUTE MANAGER

ERROR 90348

VMON-CLI

SERVICE-CONTROL

VCF 9.0

VMware Cloud Foundation

vCenter vmware-certificatemanagement Service Recovery

Prepared by: Virtual Control LLC **Date:** May 6, 2026 **Document Version:** 1.0 **Classification:** Internal — Recovery Procedure **Scope:** VCF 9.0.1 nested lab, NSX Manager Compute Manager registration

1. Executive Summary

Where commands run in this recovery doc. Recovery touches:

- **vCenter shell (SSH to VCSA root)** — `service-control --start vmware-certificatemanagement , service-control --status , log inspection under /var/log/vmware/certificate-management/ .`
- **NSX Manager UI (browser)** — Compute Manager registration retry at System > Fabric > Compute Managers.
- **NSX Manager API (curl)** — for scripted compute-manager registration verification.

Replace any credential placeholder like `<your-password>` with values from your vault.

When registering vCenter as a Compute Manager on NSX 9, registration may fail with **error code 90348**: *"Failed to import the trusted root certificate for compute manager <vcenter-ip>. Try again."* The error message implies a transient retry-able condition, but in the case documented here the retry never succeeded across 30+ minutes of attempts.

The actual root cause was that vCenter's `vmware-certificatemanagement` vmon-managed service was in **STOPPED** state. NSX's CM register flow calls vCenter's `/api/vcenter/certificate-management/vcenter/trusted-root-chains` REST endpoint to fetch the trusted root chain to import into NSX's local trust store. When `vmware-certificatemanagement` is stopped, that endpoint returns **HTTP 503 Service Unavailable** with payload `com.vmware.vapi.endpoint.cis.ServiceUnavailable`. NSX wraps this as a generic Java RuntimeException and surfaces the user-facing message above.

The fix is to start `vmware-certificatemanagement` via `service-control --start` on the vCenter appliance shell. After a brief (~15 second) initialization, the cert-mgmt endpoint returns HTTP 200 and NSX CM register succeeds on the next attempt. This was verified end-to-end during a 2026-05-06 NSX rebuild incident.

This document captures the diagnostic flow that distinguished the real cause from the misleading error message, the verified fix procedure, and the operational guardrails that prevent this from blocking future recoveries.

2. Symptoms

2.1 NSX-side symptoms

When attempting to register a Compute Manager either via the NSX UI **System → Fabric → Compute Managers → + Add Compute Manager** workflow or via the API `POST /api/v1/fabric/compute-managers`, the registration fails

with:

```
{
  "httpStatus" : "BAD_REQUEST",
  "error_code" : 90348,
  "module_name" : "inventory-mgmt",
  "error_message" : "Failed to import the trusted root certificate for compute manager 192.168.1.69. Try again."
}
HTTP 400
```

The UI shows a red banner at the top of the form with the same message. Retries fail identically.

2.2 NSX log signature

In `/var/log/proton/cm-inventory/cm-inventory.log` (or the consolidated `/var/log/syslog`):

```
ERROR http-nio-127.0.0.1-7443-exec-2 VcUtilsImpl 79635 SYSTEM [...errorCode="MP90348"...]
Failed to import Trusted Root certificate for Compute Manager : 192.168.1.69 due to RunTime Exception
com.vmware.nsx.management.lcm.common.exception.LcmRestException:
org.springframework.web.client.HttpServerErrorException$ServiceUnavailable:
503 Service Unavailable: "{
  "type": "com.vmware.vapi.std.errors.service_unavailable",
  "value": {
    "error_type": "SERVICE_UNAVAILABLE",
    "messages": [{
      "args": [],
      "default_message": "Service unavailable.",
      "id": "com.vmware.vapi.endpoint.cis.ServiceUnavailable"
    }]
  }
}"
    at
com.vmware.nsx.management.lcm.common.connection.client.rest.LcmRestClient.reportException(LcmRestClient.java:186)
    at
com.vmware.nsx.management.cminv.vcplugin.VcUtilsImpl.importVcTrustedRootCertificates(VcUtilsImpl.java:640)
    at
com.vmware.nsx.management.cminv.computemanager.service.ComputeManagerServiceImpl.configureCertificate(ComputeManagerServiceImpl.java:866)
    at
com.vmware.nsx.management.cminv.computemanager.service.ComputeManagerServiceImpl.addComputeManager(ComputeManagerServiceImpl.java:587)
```

The stack trace pinpoints `importVcTrustedRootCertificates` calling vCenter and getting back HTTP 503 from the CIS endpoint.

2.3 vCenter-side symptom

The vCenter VAMI health endpoint `https://vcenter/api/appliance/health/system` may return `"orange"`, but other health categories (mem, storage, applmgmt) often look green — the cert-mgmt service being stopped doesn't always trigger an overall-health alarm.

The smoking gun is on the cert-mgmt endpoint directly:

```
$ TOKEN=$(curl -sk -X POST -u "administrator@vsphere.local:..." \
  "https://192.168.1.69/api/session" 2>/dev/null | tr -d ' ')
$ curl -sk -H "vmware-api-session-id: $TOKEN" \
  "https://192.168.1.69/api/vcenter/certificate-management/vcenter/trusted-root-chains" \
  -w "\nHTTP %{http_code}\n"
{"error_type": "SERVICE_UNAVAILABLE",
 "messages": [{"args": [], "default_message": "Service unavailable."},
```

```
"id":"com.vmware.vapi.endpoint.cis.ServiceUnavailable"]}]}
```

HTTP 503

3. Diagnosis

3.1 Confirm cert-mgmt endpoint is the failing one

Use the verified vCenter API session token (we know auth works because the same admin can log into vCenter). Hit the same endpoint NSX is hitting:

```
TOKEN=$(curl -sk -X POST -u "administrator@vsphere.local:<pw>" \
  "https://<vcenter>/api/session" 2>/dev/null | tr -d ' ')

curl -sk -H "vmware-api-session-id: $TOKEN" \
  "https://<vcenter>/api/vcenter/certificate-management/vcenter/trusted-root-chains" \
  -w "\nHTTP %{http_code}\n"
```

HTTP 200 + JSON list = endpoint healthy, look elsewhere for the NSX failure.

HTTP 503 + `com.vmware.vapi.endpoint.cis.ServiceUnavailable` = cert-mgmt service is down. Continue with this fix.

3.2 Enable bash shell on vCenter

vCenter SSH lands in `appliance$` (PI shell), not bash. Most diagnostic commands need bash, so enable the shell mode first:

```
TOKEN=$(curl -sk -X POST -u "administrator@vsphere.local:<pw>" \
  "https://<vcenter>/api/session" 2>/dev/null | tr -d ' ')

# Enable bash for next 60 min
curl -sk -H "vmware-api-session-id: $TOKEN" -H "Content-Type: application/json" \
  -X PUT "https://<vcenter>/api/appliance/access/shell" \
  -d '{"enabled":true,"timeout":3600}' -w "\nHTTP %{http_code}\n"
# Expected: HTTP 204
```

3.3 Use `pi shell` to run commands non-interactively

Once bash is enabled, the `appliance$ pi shell <cmd>` syntax runs a bash command and exits. This avoids the interactive `shell` → bash prompt loop that hangs `plink -batch`.

```
plink -ssh -batch -pw '<vc-root-pw>' root@<vcenter> 'pi shell vmon-cli -l'
# Lists all vmon-managed services with state: Running or Stopped
```

In the 2026-05-06 incident, the output included:

```
Running:
applmgmt lookupsvc lwsmd observability pschealth vc-ws1a-broker vlcm vmafdd vmcad
vmdird vmware-analytics vmware-certificateauthority vmware-cis-license vmware-content-library
vmware-eam vmware-envoy vmware-envoy-hgw vmware-envoy-sidecar vmware-envoy-system-proxy
vmware-infraprofile vmware-perfcharts vmware-postgres-archiver vmware-rhttpproxy vmware-sca
vmware-sps vmware-std vmware-topologysvc vmware-trustmanagement vmware-updatemgr
vmware-vapi-endpoint vmware-vdtc vmware-vmon vmware-vpostgres vmware-vpxd vmware-vpxd-svcs
vmware-vsan-health vmware-vsm vsphere-ui vstats vtsdb wcp
```

Stopped:

```
observability-vapi vmcam vmware-certificatemanagement vmware-hvc vmware-imagebuilder  
vmware-netdumper vmware-rbd-watchdog vmware-vcha
```

vmware-certificatemanagement is in the Stopped list. **This is the diagnostic confirmation.**

Note on the other Stopped services: vmware-vcha, vmcam, vmware-imagebuilder, vmware-netdumper, vmware-rbd-watchdog, and vmware-hvc are intentionally stopped on a single-instance vCenter without those features enabled. These do NOT need to be started. Only vmware-certificatemanagement is the recovery target.

4. The Fix — Start vmware-certificatemanagement

4.1 One-shot start command

```
plink -ssh -batch -pw '<vc-root-pw>' root@<vcenter> \  
'pi shell service-control --start vmware-certificatemanagement'
```

Expected output:

```
Operation not cancellable. Please wait for it to finish...  
Performing start operation on service certificatemangement...  
Successfully started service certificatemangement  
Shell access is granted to root
```

4.2 Verify the service is running

```
plink -ssh -batch -pw '<vc-root-pw>' root@<vcenter> \  
'pi shell service-control --status vmware-certificatemanagement'
```

Expected:

```
Running:  
vmware-certificatemanagement
```

4.3 Verify the cert-mgmt endpoint now returns 200

Wait ~15-30 seconds for the service to fully initialize, then retry the cert-mgmt API call:

```
TOKEN=$(curl -sk -X POST -u "administrator@vsphere.local:<pw>" \  
  "https://<vcenter>/api/session" 2>/dev/null | tr -d ' ')  
  
for i in 1 2 3 4 5; do  
  sleep 15  
  http=$(curl -sk --max-time 10 -H "vmware-api-session-id: $TOKEN" \  
    "https://<vcenter>/api/vcenter/certificate-management/vcenter/trusted-root-chains" \  
    -o /tmp/resp.txt -w "%{http_code}" 2>&1)  
  echo "[t+${i}*15s] HTTP $http"  
  if [ "$http" = "200" ]; then  
    echo "endpoint OK"  
    head -c 300 /tmp/resp.txt  
    break  
  fi  
done
```

```
fi
done
```

Expected:

```
[t+1*15s] HTTP 200
endpoint OK
[{"chain":"6E22DB1BC54ADF67BF052C1B4BB26B3F76CD2B16"}]
```

4.4 Retry NSX CM register

With the cert endpoint now healthy, NSX CM register works on the first attempt:

```
THUMB=$(echo | openssl s_client -connect <vcenter>:443 -servername <vcenter> 2>/dev/null \
| openssl x509 -fingerprint -sha256 -noout | sed 's/.*=//')

cat > /tmp/cm-register.json <<EOF
{
  "server": "<vcenter-ip>",
  "origin_type": "vCenter",
  "credential": {
    "credential_type": "UsernamePasswordLoginCredential",
    "username": "administrator@vsphere.local",
    "password": "<pw>",
    "thumbprint": "$THUMB"
  },
  "create_service_account": true,
  "set_as_oidc_provider": true,
  "access_level_for_oidc": "FULL"
}
EOF

curl -sk --tlsv1.3 -u "admin:<nsx-pw>" \
-X POST "https://<nsx-mgr>/api/v1/fabric/compute-managers" \
-H "Content-Type: application/json" \
-d @/tmp/cm-register.json -w "\nHTTP %{http_code}\n"
```

Expected:

```
{
  "server" : "<vcenter-ip>",
  "origin_type" : "vCenter",
  ...
  "id" : "<new-cm-uuid>"
}
HTTP 201
```

After ~60 seconds, verify CM status is Up + Registered:

```
CM_ID="<new-cm-uuid>"
curl -sk -u "admin:<nsx-pw>" "https://<nsx-mgr>/api/v1/fabric/compute-managers/$CM_ID/status"
# Expected: connection_status: UP, registration_status: REGISTERED
```

5. Step-by-Step from the 2026-05-06 Incident

This is the verbatim sequence that resolved the issue.

Time UTC	Event
~13:43	New NSX Manager appliance (192.168.1.71) booted, cluster STABLE/STABLE
~13:50	First CM register attempt — HTTP 400 error 90344 "Enable Trust must be true"
~13:51	Adjusted JSON: <code>set_as_oidc_provider: true</code> , retry
~13:52	Second attempt — HTTP 400 error 90348 "Failed to import the trusted root certificate for compute manager 192.168.1.69. Try again."
13:53-14:30	8+ retries spaced 30s apart — all HTTP 400 / error 90348
~14:35	Began investigating: tried passing explicit <code>extension_certificate.pem_encoded</code> — rejected ("private_key required")
~14:40	Direct probe of vCenter endpoint with auth token → HTTP 503 Service Unavailable
~14:45	Enabled bash shell on vCenter via <code>PUT /api/appliance/access/shell {"enabled":true,"timeout":3600}</code>
~14:48	<code>pi shell vmon-cli -l</code> → <code>vmware-certificatemanagement</code> listed under Stopped
14:50	<code>pi shell service-control --start vmware-certificatemanagement</code> → Successfully started
14:50:15	First retry of cert-mgmt endpoint → HTTP 200 with chain payload
14:51	NSX CM register POST → HTTP 201 created
14:52	CM status → <code>connection_status: UP, registration_status: REGISTERED</code>

Total recovery time after correct diagnosis: **~5 minutes** (mostly waiting for the service to initialize and the cert-mgmt sync to settle).

Time wasted on misleading "Try again" retries before correct diagnosis: **~40 minutes**.

6. Why the Service Stops in the First Place

6.1 Observed correlation

The `vmware-certificatemanagement` service has been observed in stopped state after:

- Hard power-off of the vCenter VM (e.g., during a workstation outage or panic shutdown of the host)
- vCenter VM under sustained memory pressure (VAMI health "orange" for mem)
- Long uptime without restart (multi-month uptime sometimes correlates with vmon services drifting to stopped state)

6.2 vmware-certificatemanagement does NOT auto-restart

Unlike some critical vCenter services (`vmware-vpxd`, `vmware-vapi-endpoint`, `vmware-stds`) which vmon will retry indefinitely, `vmware-certificatemanagement` has a **finite retry count** before vmon gives up and marks it permanently Stopped. Once stopped, manual `service-control --start` is required.

6.3 Detection-only-when-needed problem

Because cert-mgmt is only consumed by relatively rare operations (CM register, cert renewal, identity broker setup), a stopped service can sit for weeks without anyone noticing. Then one day a CM register fails with the misleading

"try again" error and the operator chases the wrong cause.

7. Lessons Learned

#	Lesson
L-1	NSX error 90348 "Try again" is a misleading message. The retry suggestion implies transient condition; in reality the cert-mgmt service is permanently stopped on vCenter and requires manual restart. Do not retry indefinitely.
L-2	Always probe the vCenter endpoint NSX is calling before retrying. One curl call (/api/vcenter/certificate-management/vcenter/trusted-root-chains) returns HTTP 503 immediately, distinguishing service-down from auth/network issues.
L-3	Use <code>pi shell vmon-cli -l</code> to enumerate vmon services. The output groups them under Running and Stopped. Compare against the expected baseline for a single-instance vCenter to spot anomalies.
L-4	Bash access on vCenter is required for vmon and service-control commands. Enable via the REST API: <code>PUT /api/appliance/access/shell {"enabled":true,"timeout":3600}</code> — no SSH needed for the enable step.
L-5	After fixing the cert-mgmt service, the NSX CM register works on the next attempt — no NSX-side cleanup or reconfiguration needed. The error was 100% on the vCenter side.
L-6	Other vmon services in Stopped state on a single-instance vCenter are typically intentional (vcha, vmcam, hvc, imagebuilder, netdumper, rbd-watchdog). Don't reflexively start them. Only <code>vmware-certificatemanagement</code> should be started in this scenario.

8. Operational Guardrails

8.1 Pre-flight check before NSX rebuild

When planning an NSX rebuild that will require CM re-registration:

```
# As part of pre-flight, verify cert-mgmt is healthy
TOKEN=$(curl -sk -X POST -u "administrator@vsphere.local:pw" \
  "https://<vcenter>/api/session" 2>/dev/null | tr -d '\n')

curl -sk -H "vmware-api-session-id: $TOKEN" \
  "https://<vcenter>/api/vcenter/certificate-management/vcenter/trusted-root-chains" \
  -w "\nHTTP %{http_code}\n"
# Must be HTTP 200. If 503, fix per &sect;4 BEFORE starting the NSX rebuild.
```

8.2 After any vCenter hard-off or VAMI orange health

Run the vmon enumeration explicitly:

```
plink -ssh -batch -pw '<vc-root-pw>' root@<vcenter> 'pi shell vmon-cli -l'
```

If `vmware-certificatemanagement` appears in Stopped, start it before any cert-touching operation.

8.3 Periodic check (optional)

A simple cron job on a workstation can catch this over time:

```
# Every hour: probe cert-mgmt, alert if 503
0 * * * * curl -sk -H "vmware-api-session-id: $(get_session_token)" \
  "https://<vcenter>/api/vcenter/certificate-management/vcenter/trusted-root-chains" \
  -o /dev/null -w "%{http_code}" \
  | grep -v "^200$" && echo "ALERT: vCenter cert-mgmt service down" | mail -s "vc-alert" you@example
```

9. Cross-References

- `NSX-Manager-Cold-Start-Troubleshooting-Report.md` Addendum D — the broader NSX rebuild context where this fix was discovered
- VMware KB articles on `vmware-certificatemanagement` (search Broadcom KB portal for current article ID)
- `feedback_no_assumptions.md` — the rule that drove probing the endpoint directly rather than trusting NSX's "try again" message
- `feedback_kb_prescribed_path.md` — following the chain "NSX error" → "vCenter endpoint" → "vmon service" rather than substituting heuristics

10. Document Information

Field	Value
Document Title	vCenter vmware-certificatemanagement Service Recovery
Prepared By	Virtual Control LLC
Date	May 6, 2026
Document Version	1.0
Classification	Internal — Recovery Procedure
VCF Version	9.0.1
vCenter Build Tested	9.0.1.0 build 24957454
NSX Build Tested	9.0.1.0 build 24952114
Recovery Time	~5 min after correct diagnosis (vs 40+ min on misleading retry path)